

2024



Groupama • FDJ
ÉQUIPE CYCLISTE

Innovating Professional Cycling Design on AWS with Équipe cycliste Groupama-FDJ

Learn how cycling team Équipe cycliste Groupama-FDJ accelerated bike and equipment design using Ansys Gateway powered by AWS.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#) | [Architecture Diagram](#)

17x faster simulations

on AWS versus on premises



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2

26 hours reduced to 1.5 hours

for a single global simulation

Overview

In professional cycling events such as the Tour de France, a single second can make or break the outcome of a race. To gain a competitive edge and win more races, French professional cycling team [Équipe cycliste Groupama-FDJ](#) uses computational fluid dynamics (CFD) and computer assisted design (CAD) to optimize the aerodynamics of their bikes, helmets, and other equipment.

Since 2021, the team has used software from [Ansys](#), an [AWS Partner](#) and engineering simulation solutions provider, to run CFD simulations using [Ansys Fluent](#). Équipe cycliste Groupama-FDJ is also using other Ansys tools, including [Ansys SpaceClaim](#) and [Ansys Discovery](#), to design parts using CAD and to reveal critical insights early in the design process.

As the team's simulations grew to include entire rider and bicycle systems, its on-premises computing power could no longer keep pace. To accelerate its CFD processes, Équipe cycliste Groupama-FDJ began to run these workloads on Amazon Web Services (AWS). Now, the team can run faster simulations on AWS and explore even more complex configurations and uncover new insights to boost performance.



Opportunity | Using Ansys Gateway Powered by AWS Accelerate Bike Development for Équipe cycliste Groupama-FDJ

Founded in 1997, Équipe cycliste Groupama-FDJ competes at the UCI WorldTeam level. The team has won nearly 550 victories and achieved 48 national champion jerseys, 3 monuments of cycling, and 38 Grand Tour stages. To maintain a competitive edge, its research and development unit tests new bicycle designs and technologies using multiple Ansys tools, such as Ansys Fluent, a fluid simulation software known for its advanced physics modeling capabilities.

Through CFD simulations, Équipe cycliste Groupama-FDJ can analyze the complex interactions between the rider, bicycle, and surrounding air. With these insights, the team can make changes to their equipment to help improve a rider's performance.

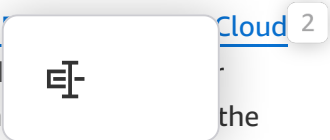
Previously, Équipe cycliste Groupama-FDJ used a small, specially configured computer to run CFD simulations on premises. "We wanted to better understand how different factors, such as speed and side wind, affect the performance of various bicycle components like wheels and helmets," says Victor Simonin, research and development engineer at Équipe cycliste Groupama-FDJ. "However, with our previous computer setup, the time required to run even a single simulation was too long, which limited our ability to develop and test new parts."

To address this challenge, Équipe cycliste Groupama-FDJ looked to the cloud and discovered [Ansys Gateway powered by AWS](#), a solution for developers, designers, and engineers who want to manage their complete Ansys workflows in the cloud. Using this solution, the team could quickly set up a high-performance computing environment on AWS to deploy Ansys Fluent and run CFD simulations faster than ever before.

Solution | Running Simulations on AWS 17 Times Faster Compared to On-Premises Technology

In late 2023, Équipe cycliste Groupama-FDJ established an AWS account and migrated to Ansys Gateway powered by AWS. The implementation process was seamless because the cloud-based solution provided a familiar Ansys software interface, requiring minimal IT expertise from the team. Because the team was able to procure the solution through [AWS Marketplace](#)—where organizations can discover, deploy, and manage software that runs on AWS—it was able to quickly access and deploy the necessary tools and resources. In mid-January 2024, the team successfully ran its first CFD simulation on AWS, even without having prior experience with AWS.

Équipe cycliste Groupama-FDJ gained access to powerful compute resources such as [Amazon EC2 Hpc7a Instances](#), which are designed to run tightly coupled HPC workloads such as forecasting, and multiphysics simulations with better performance. (See figure 1 below.) With the team has expanded its testing capabilities. It can now test 4–5 prototypes at the same time, compared to only 2

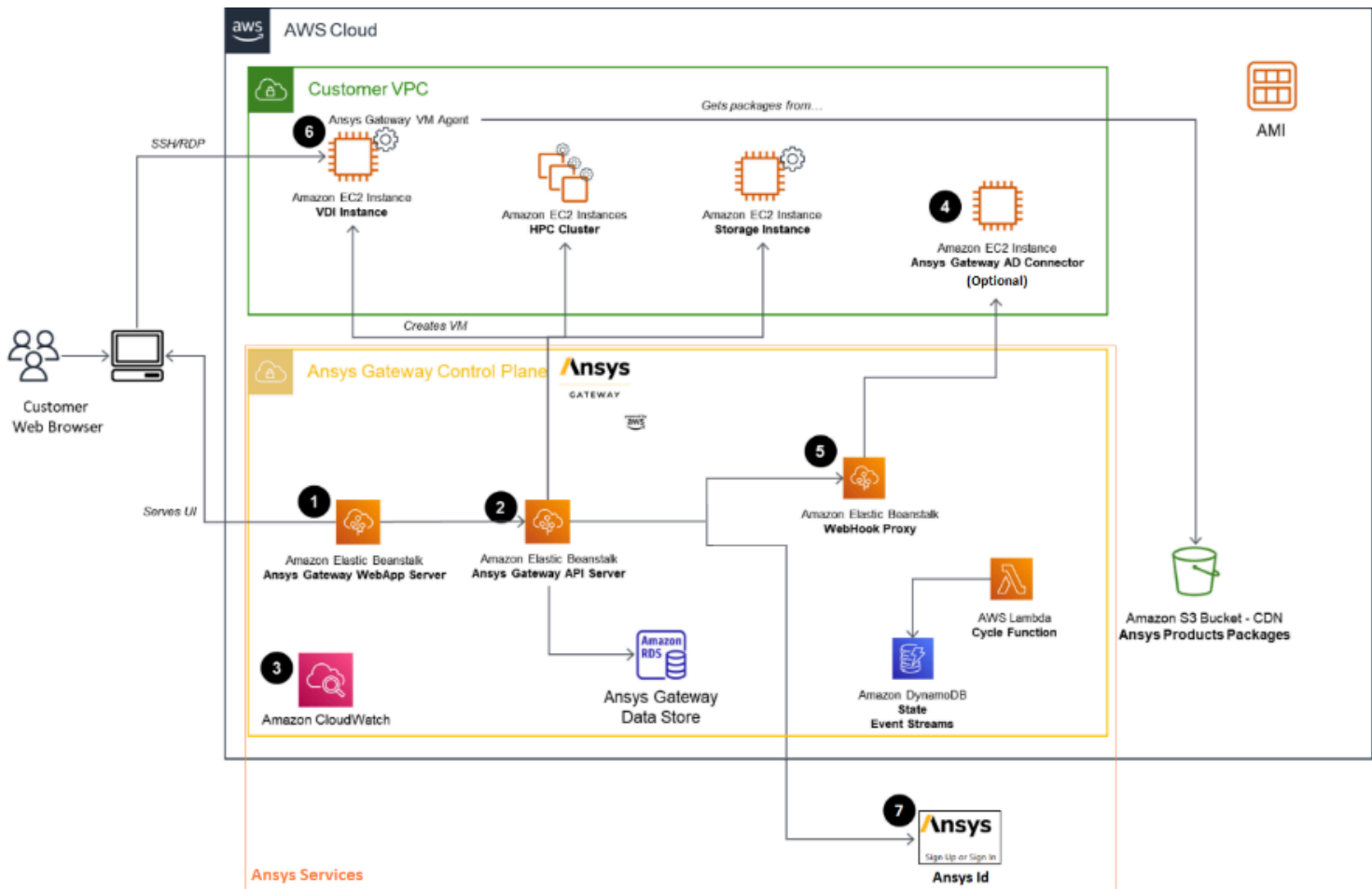


prototypes on premises. With this newfound efficiency, the team can explore a wider range of design possibilities and iterate more quickly. The team can run 2–3 times more simulations per day, covering a broader range of design parameters and scenarios.

“We tested a complete system, including the rider, helmet, and wheels, using Ansys Gateway powered by AWS,” says Victor. “Previously, we could only afford to perform this test a few times due to the lengthy computation time. A single global simulation would take around 26 hours. Now, on AWS, that same simulation takes just an hour and a half, which is about 17 times faster.”

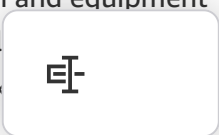
With these capabilities, Équipe cycliste Groupama-FDJ can now innovate and optimize its cycling equipment at a much faster pace. It can also run more complex simulations thanks to the increased computing power, uncovering new insights. For example, using Ansys Discovery, it quickly performed testing and validation during the ideation phase, which accelerated the process that was completed later with CFD simulations. It also used Ansys SpaceClaim to design handlebars using CAD. With these advancements, the team can develop more efficient equipment and help its cyclists perform their best on the track.

Architecture Diagram



Outcome | Launching New Prototypes to Obtain a Competitive Edge

With Ansys Gateway powered by AWS, Équipe cycliste Groupama-FDJ has the potential to expand its CFD simulation capabilities even further. It plans to explore simulations that involve multiple team and equipment configurations to optimize performance at a larger scale. The team will also use the cloud to launch new prototypes, incorporate data from more riders, and, ultimately, boost its standing in high-stakes cycling competitions.



“Our work with Ansys and the AWS team will certainly start the development of a new bike,” says Fred Grappe, director of performance at Équipe cycliste Groupama-FDJ. “We also plan to improve our helmets with this technology. And by integrating more riders on the team, we will be able to better understand their aerodynamics. This will not only increase the capacity of our technology but also increase our competitive edge.”

About Équipe cycliste Groupama-FDJ

Équipe cycliste Groupama-FDJ is a French professional cycling team that competes at the UCI WorldTeam level. Founded in 1997, the team has won nearly 550 victories and 38 Grand Tour stages.

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